Woojung Song

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RESEARCH INTERESTS

I study how LLM agents behave and how to make them more reliable and better aligned with human needs and values. My work spans behavioral evaluation, value measurement, tool use, and user interaction. I am also interested in social agents that model the values, personalities, and changing behavior of people and fictional characters.

EDUCATION

Seoul National University, Seoul, South Korea — Sep 2025 – Present
Integrated M.S.–Ph.D., Graduate School of Data Science
Advisor: Yohan Jo, HOLI Lab (Human-Oriented Language Intelligence)

Hanyang University, Seoul, South Korea — Mar 2022 – Aug 2025
B.I.S. Department of Information Systems — Overall GPA: 4.35/4.5 | Major GPA: 4.35/4.5

PUBLICATIONS

* Equal contribution

Source Preference in the Wild: How LLM Agents Favor Items by Source, and How to Reduce It
Under Review 2026
Jonghyun Song*, Haewon Park*, Jeonghoon Shim*, **Woojung Song**, Yohan Jo

AgentHabit: Characterizing Distinct Behaviors of Agents on Everyday Tasks
Under Review 2026
Woojung Song*, Hoyeol Yang*, Jeonghoon Shim, Sungjib Lim, Jonggeun Lee, Yunho Choi, Yohan Jo

Agents' Overreliance on Unreliable Tools
Under Review 2026
Hoyeol Yang*, **Woojung Song***, Taewon Kim, Jonghyun Song, Seoyeon Park, Yohan Jo

ArcANE: Do Role-Playing Language Agents Stay in Character at the Right Time?
Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), Main 2026
Woojung Song*, Nalim Kim*, Sangjun Song, Chaewon Heo, Jongwon Lim, Yohan Jo

Human Psychometric Questionnaires Mischaracterize LLM Behavior
Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), Main 2026
Woojung Song*, Dongmin Choi*, Yoonah Park, Jongwook Han, Eun-Ju Lee, Yohan Jo

Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL), Main 2026
Jonggeun Lee*, **Woojung Song***, Jongwook Han, Haesung Pyun, Yohan Jo

Non-Collaborative User Simulators for Tool Agents
The Fourteenth International Conference on Learning Representations (ICLR) 2026
Jeonghoon Shim, **Woojung Song**, Cheyon Jin, Seungwon Kook, Yohan Jo

Psychometric Item Validation Using Virtual Respondents with Trait-Response Mediators
Transactions of the Association for Computational Linguistics (TACL) 2026
Sungjib Lim, **Woojung Song**, Eun-Ju Lee, Yohan Jo

Quantifying Data Contamination in Psychometric Evaluations of LLMs
Findings of the Association for Computational Linguistics: EACL 2026
Jongwook Han*, **Woojung Song***, Jonggeun Lee*, Yohan Jo

Value Portrait: Assessing Language Models' Values through Psychometrically and Ecologically Valid Items
Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Main 2025
Jongwook Han*, Dongmin Choi*, **Woojung Song***, Eun-Ju Lee, Yohan Jo

Interpretable Prediction of Private Brand Purchases by Pet Type in E-Commerce for Consumer Behavior Analysis Using Real-World Transaction Data
PeerJ Computer Science 2026
Jaehyuk Lee*, **Woojung Song***, Jina Kim, Yoona Chung, Eunchan Kim

RESEARCH EXPERIENCE

Seoul National University, HOLI Lab — Seoul, South Korea
Graduate Research, Integrated M.S.–Ph.D. Program — Sep 2025 – Present
Undergraduate Research Intern — Jul 2024 – Aug 2025

- Characterize how agents behave across everyday tasks and respond to unreliable tools (AgentHabit; Agents' Over-reliance on Unreliable Tools).
- Develop and evaluate methods for measuring LLM values and assessing consistency between questionnaire responses and observed behavior.
- Study how role-playing agents track changes in fictional characters' values, personalities, and behavior over time (ArcANE).

WORK EXPERIENCE

LG Electronics, H&A Sound/Image Intelligence R&D Lab — Seoul, South Korea
Intern — Mar 2024 – Jun 2024

- Enhanced LLM-based Web Question Answering using RAG and multi-agent approaches
- Developed WAAG, an automated QA evaluation system with 40% accuracy improvement
- Achieved highest level in LG SW Competency Certification Test

US 8th Army, Korean Augmentation To the United States Army — Pyeongtaek, South Korea
Sergeant (Senior KATUSA) — May 2020 – Nov 2021

- Served in US 8th Army as part of the mandatory military service program
- Received Certificate of Appreciation for exceptional service

SCHOLARSHIPS

Presidential Science Scholarship for Graduate Students — South Korea
Ministry of Science and ICT — 2026 – 2027

AWARDS

Busan Mayor's Award — Busan, South Korea
DIVE Competition, 2nd Place — 2026

- Also won 1st Place from Busan Metropolitan Corporation in the same competition

Chuncheon Mayor's Award — Chuncheon, South Korea
Chuncheon Generative AI Competition, 3rd Place — Jun 2025

- Developed "Blue-Ming" Chuncheon City Youth Entrepreneurship AI Consulting Platform

Busan Mayor's Award — Busan, South Korea
DIVE Competition, App and Service Development, 3rd Place — 2024

- Developed "Korail Donghae" application integrating Language Models with custom tools and RAG
- Also won 1st Place from Korea Railroad Corporation Busan-Gyeongnam Headquarters